IERG4320 Data Science in Practice (2025-26 Term 1)
Assignment 1
Expected time: 6 hours

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Learning outcomes:
1. To practise basic Python programming and basic statistics. 2. To obtain hands-on experience with `NumPy`, `Pandas` and `SciPy` packages. 3. To understand the basic concepts of a data science project.

Instructions:

1. Do your own work. You are welcome to discuss the problems with your fellow classmates. Sharing ideas is great, and do write your own explanations.
2. If you use help from the AI tools, e.g. ChatGPT, DeepSeek and Gemini, write clearly how much you obtain help from the AI tools. No marks will be taken away for using any AI tools with a clear declaration.
3. All work should be submitted onto the blackboard before the due date.
4. You are advised to submit the following two items.
   a. A .pdf file containing the answers of the written part.
   b. A Assignment1_1155xxxxxx.py file storing all your programs for the five problems. (1155xxxxxx refers to your student ID)
5. Do type/write your work neatly. If we cannot read your work, we cannot grade your work.
6. We will grade your work based on Anaconda Python version 1.13.
7. The sample output in the specification is run based on Windows. Your output may be different if you run the programs on Mac or Linux machines.
8. If you do not put down your name, student ID in your submission (both .pdf and .py), you will receive a 10% mark penalty out of the assignment 1.
9. Due date: 20 September, 2025 (Saturaday）23:59

Short questions (25%)
Answer all questions.

1. You are working for data engineering in the Hong Kong Innovation and Technology Commission. The Hong Kong government is planning to develop some AI models for internal use. Your boss asks you something about Ethical Artificial Intelligence Framework:

    a. What department drafts the Ethical Artificial Intelligence Framework? (1%)

    b. What are the **four related laws or regulations** relevant to harmful artificial intelligence practices? (4%)

    c. The immigration department uses a facial recognition system to detect human faces in the express E-channel service. What is the level of risk of this model? Explain your choice. (2%)

    d. The transport department uses a car recognition system to monitor the traffic on the road. What is the level of risk of this model? Explain your choice. (2%)

    e. Suppose that the Education Bureau trains an AI teacher using generative AI. This model assists the students in solving high school mathematics problems. What is the level of risk of this model? Explain your choice. (2%)

    f. Suppose that the Hospital Authority trains an AI assistant using deep learning. This model acts as the triage system to set the priority categories of the patients in the Accident and Emergency (A&E). What is the level of risk of this model? Explain your choice. (2%)

    g. After studying the examples from part c to part f, suggest **three more examples** on the use of AI with **different departments** in the Hong Kong Government. In addition, **state and explain the level of risk** of your examples. (9%)

    h. In the Ethical AI Principles, what are transparency and Interpretability? Why are transparency and Interpretability important? How can transparency and Interpretability be achieved? (3%)

(1a) Digital Policy Office

(1b) Privacy Transparency
Bias prevention Accountability

(1c) High Risk
Since it is related to AI for record-keeping
With the process and collection of biometric data
will lead to serious privacy violations

(1d.) Limited Risk
Since it only includes information of car
(X biometric data / X affect drivers)

(1e.) Limited Risk
Since AI teaching only use to educate student
there are some teaching sources only (X information)

(1f.) High Risk
Since deep learning in nosipital is related to human
life and incidents

(1g.)

1. Labour Department (limited Risk)
   - for recruitment screening affect only the job opportunity
2. Environmental Protection Department (Low Risk)
   - for public information only which do not affect individual rights
3. Social Welfare Department (High Risk)
   - Analyzes application data to evaluate CSSA is related to fundamental right and unfair treatment

(1h.) Transparency:

Design / operation and decision making process of AI-system are clearly visible to both users and regulators

Interpretability:

Decision made by AI can be understood by human and it could be explained.

(Q2) They are important since the user could trust that the AI system is telling truth, eventhrough AI is wrong, they still have record to show and follow the wrong case, to ensure the result are fair to everyone

(Q3) Achieved by using interpretable models, providing decision Rationales, Log and Audit and Human-centered interface

Practical problems (75%)
Work on the following five problems in a single .py file.
If your Python program cannot be run, you will receive no scores for the problem.
Put down your student ID as a comment in your first line of the program.

2. In the data file, it contains a matrix of numbers. Write a function that evaluates (i) row mean, (ii) row standard deviation, (iii) column sample mean, and (iv) column sample error based on the input dataframe (a matrix of numbers). (15%)
   This problem covers basic statistics.

   Code:

```
# <Your student ID>
import numpy as np
import pandas as pd
# Problem 2
def problem_2(n):
    row_mean = row_sd = column_sample_mean = column_ss = 0
    # write your logic here

    return row_mean, row_sd, column_sample_mean, column_ss
```

   Execution:

```
> df = pd.read_csv('problem2.csv', sep=',', header=None)
> print(problem_2(df))
(0     6.166667
1    12.166667
2     5.000000
3     9.333333
4    10.166667
5     9.500000
6     7.833333
7     9.500000
dtype: float64, 0    3.488075
1    6.431692
2    4.049691
3    6.088240
4    4.355074
5    1.974842
6    6.369197
7    3.016621
dtype: float64, 0     7.375
1     5.875
2     9.000
3     9.375
4    11.625
5     9.000
dtype: float64, 0    1.534572
1    1.315261
2    1.945691
3    1.522891
```

```
4    1.534572
5    2.171241
dtype: float64)
```

problem2.csv:

```
5,1,11,9,6,5
8,8,5,13,17,22
2,3,4,11,9,1
15,3,2,17,10,9
8,7,18,6,11,11
9,12,8,8,12,8
2,4,9,3,19,10
10,9,15,8,9,6
```

3. Write a function to calculate the p-value and Chi-square statistics from the Chi-squared tests based on a list of the sum of two special dice. The faces of the special dice are {1, 1, 2, 3, 4, 4}. The purpose of the test is to see if the two dice are fair dice based on the observed sum. You can assume that all sums are between 2 and 8 inclusively. (15%) This problem covers basic statistics on Chi-squared tests.

Code:

```
# Problem 3
from scipy.stats import chisquare
def problem_3(list_of_observation):
    p = chi2 = 0
    # write your logic here

    return p, chi2
```

Execution:

```
> observation = [3,5,3,6,7,8,3,5,5,2,4,4,5,2,8,7,5,5,5,3]
> p, chi2 = problem_3(observation)
> print("p-value :", p)
p-value : 0.7807710149582182
> print("chi-square :", chi2)
chi-square : 3.2199999999999998
```

4. Write a function that evaluates (i) the p-value from the paired sample T-test and (ii) the p-value from the Wilcoxon signed-rank test with T-statistics based on a list of paired samples. (15%)
This problem covers basic statistics on paired sample T-test and Wilcoxon signed-rank test.

Code:

```
# Problem 4
from scipy.stats import wilcoxon, ttest_rel
def problem_4(df):
    pairt = w = 0
    # write your logic here

    return pairt, w
```

Execution:

```
> df = pd.read_csv('problem4.csv', sep=',', header=None)
> pairt, w = problem_4(df)
> print("p-value from paired sample T-test: ", pairt)
p-value from paired sample T-test:  0.19747335562857904
> print("p-value from wilcoxon signed-ranked test with T-statistics:
", w)
p-value from wilcoxon signed-ranked test with T-statistics:
0.35546875
```

problem4.csv:

```
60,61
70,71
40,38
41,39
40,38
40,33
45,55
48,56
30,38
50,68
```

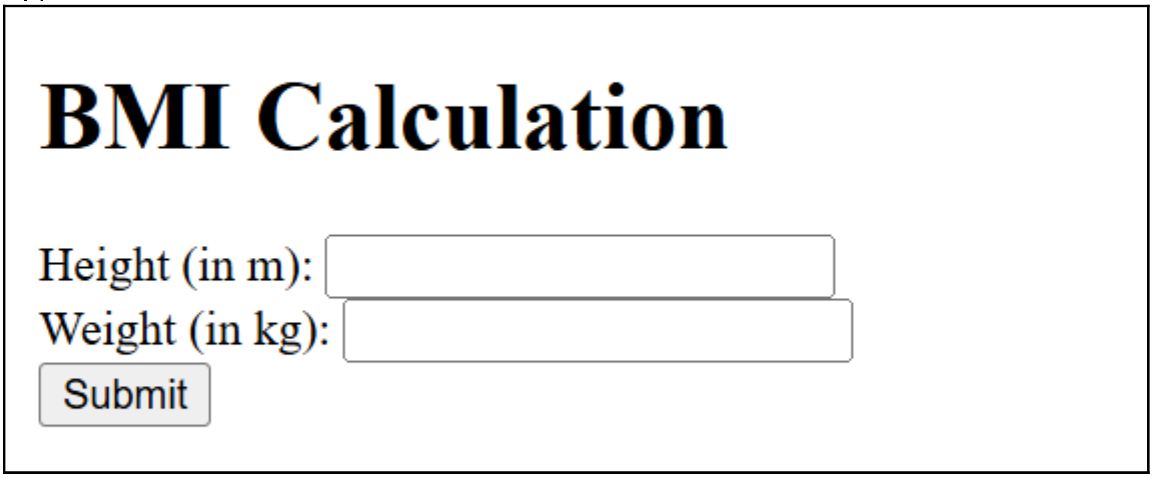
5. Design your html form having the following properties. (15%)
    - The title of the application is “BMI Calculation”
    - A form having the field “height” and “weight”
    - “height” is a floating point number with minimal value 0 and maximum value 2.5.
    - “weight” is another floating point number with minimal value 0 and maximum value 150.
    - The form submits the data to the path `/model`, using post-method.

    This problem covers a basic web application in Python.

    Code:

```html
<!-- Problem 5 -->
<!DOCTYPE html>
<html>
   <head>
      <meta charset="utf-8">
      <meta name="viewport" content="width=device-width,
initial-scale=1">
   </head>
   <body>
      <!-- Fill in your form here -->
   </body>
</html>
```

    Appearance:



6. Design your simple FastAPI server with the following properties: (15%)
   - If height is zero, return a JSON response, `{"result":-1.0,"error":"Height is zero"}`
   - If height is not zero, return another JSON response, `{"result":x,"error":"Height is zero"}`, where x is the BMI.
   - Other errors checking should be done in the frontend (e.g. negative weight, super tall human, … etc)

   This problem covers a basic FastAPI server in Python.

   Code:

```
# Problem 6
# start cmd: uvicorn problem6:app --host 0.0.0.0 --port 55726
# test at: http://localhost:55726/docs
from typing import Annotated
from fastapi import FastAPI, Form, Request
from fastapi.responses import JSONResponse, HTMLResponse
from fastapi.encoders import jsonable_encoder
from fastapi.templating import Jinja2Templates
from pydantic import BaseModel
import pickle

app = FastAPI()

@app.post("/model/")
async def login(data: Annotated[ModelData, Form()]):
    # write your logic here

    return JSONResponse(content="", status_code=200)

templates = Jinja2Templates(directory='templates')
@app.get('/', response_class=HTMLResponse)
async def main(request: Request):
    return templates.TemplateResponse('problem5.html', {'request':
request})
```

   Execution: when height = 1.8, weight = 60

```
{"result":18.51851851851852,"error":""}
```

< End of Assignment >